

Thermal Engineering Lab

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

6439A

Credits

4

Periods

5/week

Course Details

Title: Thermal Engineering Lab**Department:** Common**Revision:** REV_Thermal Engineering Lab**Pattern:** Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6439A — Thermal Engineering Lab**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ു ഐംഗ്ലീഷ് ടെക്നിക്കൽ ടെർംസ്-ഐംഗ്ലീഷിൽ ുയ്യ്. ുയ്യ് ഐംഗ്ലീഷിൽ ുയ്യ്, formula, diagram, units, definitions ുയ്യ് ുയ്യ്.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Thermal Engineering Lab.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Thermal Engineering Lab.

Study

Detailed analysis of core principles and theoretical foundations.

Malayalam Support Note: താപനിലയുടെയും താപോളക്കളുടെയും അടിസ്ഥാനപരിചയം നേടുക. താപനിലയുടെയും താപോളക്കളുടെയും അടിസ്ഥാനപരിചയം നേടുക.

MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

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MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: ടെസ്റ്റിംഗ് സിസ്റ്റം ട്രബിൾഷൂട്ടിംഗ് സിസ്റ്റം
ഡെവലപ്മെന്റ് സിസ്റ്റം.

MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: ടെസ്റ്റിംഗ് സിസ്റ്റം ട്രബിൾഷൂട്ടിംഗ് സിസ്റ്റം
ഡെവലപ്മെന്റ് സിസ്റ്റം.

Practice Model Question Paper

Time: 3 Hours | Max Marks: 75

Part A (2 marks each)

1. Define core concepts of Thermal Engineering Lab.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.